

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103

Winter Semester Examination - Dec. - 2019

Branch: B. Pharm.

Subject: Pharmacology-II (BP503T)

Date: 20/12/2019

Semester: V

Marks: 75

Time: 3hrs

Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Multiple Choice Questions (MCQs) = 20 x 1 = 20

(Answer all the questions)

- i. _____ is fat soluble vitamin essential for formation of clotting factors.
a. Vit. A b. Vit. E c. Vit. D d. Vit. K
- ii. Aspirin inhibits _____ enzyme to produce antiplatelet action.
a. Thromboxane A₂ b. Cyclooxygenase-1
c. Cyclooxygenase-2 d. Phosphodiesterase
- iii. Radioactive iodine is indicated in _____.
a. Hypothyroidism b. Addison's disease c. Hyperthyroidism d. None of the above
- iv. Active form of vitamin D is _____.
a. Cholecalciferol b. 7-dehydrocholesterol c. Calcifediol d. Calcitriol
- v. Furosemide acts by inhibiting the following in the renal tubular cell.
a. Na⁺-K⁺-2Cl⁻ cotransporter b. Na⁺-H⁺ antiporter
c. Na⁺-Cl⁻ symporter d. Na⁺ K⁺ ATPase
- vi. _____ is HMG-CoA reductase inhibitor.
a. Fenofibrate b. Nifedipine
c. Rosuvastatin d. Enalapril
- vii. Hypovolemic shock may be treated with:
a. 0.9 % normal saline b. 5 % dextrose
c. Ringer lactate d. All of the above
- viii. The primary route of administration of insulin is:
a. Intradermal b. Subcutaneous c. Intramuscular d. Intravenous
- ix. Uricosuric agent used in treatment of gout is
a. Probenecid b. Febuxostat c. Allopurinol d. Metnotrexate
- x. Metformin is the only drug of the _____ class that is used clinically.
a. Biguanides b. Meglitinide analogues c. Sulfonylureas
d. Thiazolidinediones

- xi. _____ is the main mineralocorticoid steroid hormone produced by adrenal cortex in the adrenal gland.
a. Cortisol b. Prednisolone c. Aldosterone d. Dexamethasone
- xii. The following can be used to antagonise the action of heparin in case of overdose:
a. Heparan sulphate b. Dextran sulphate c. Protamine sulphate d. Ancrod
- xiii. _____ is the more potent androgen.
a. Testosterone. b. Androstenedione c. Dehydroepiandrosterone
d. Dihydrotestosterone
- xiv. Glucagon stimulates _____ process.
a. Glycogenesis b. Glycogenolysis c. Glycolysis d. Gluconeogenesis
- xv. Following drugs are Na⁺ channel blockers except
a. Quinidine b. Pracainamide c. Diltiazem d. Lidocaine
- xvi. The drug Nicorandil is _____
a. Potassium channel opener b. Calcium channel blocker c. β blocker d. α blocker
- xvii. Cyproheptadine is _____
a. Anti-Serotonergic b. Anti-Histaminergic c. Local anesthetic d. All of above
- xviii. Myxoedema and cretinism are treated by _____
a. Antithyroid drugs b. Glucagon c. Corticosteroids d. Thyroid hormones
- xix. Insulin act through _____
a. G-protein coupled receptors
b. Enzyme linked receptors
c. Receptors regulating gene expression
d. Receptors with intrinsic ion channel
- xx. _____ produces hypocalcaemic effect.
a. Calcitonin b. Vit. D c. Parathormone d. All of the above

Q. No.2 Long Answers (Answer 2 out of 3) = 2 x 10 = 20

- A. Classify drugs used in the treatment of hypertension with examples. Explain ACE inhibitors in detail.
- B. What is diabetes mellitus? Classify oral hypoglycaemic agent and write mechanism of action of insulin.
- C. Classify NSAIDs. Explain MOA and clinical uses of Aspirin.

Q. No.3 Short Answers (Answer 7 out of 9) = 7 x 5 = 35

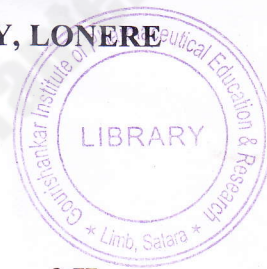
- A. Write about the synthesis, storage and release of Thyroid hormones
- B. Describe the triple response of histamine and add a note on anti-histaminics.

- C. Describe the management of hyperlipaemia.
- D. What do you mean by Bioassay? Discuss its types.
- E. What is angina pectoris. Give brief account on nitrates.
- F. Differentiate between heparin and warfarin.
- G. Classify diuretics. Discuss therapeutic uses and complications of thiazide diuretics.
- H. Write note on plasma volume expanders.
- I. Write uses and adverse effects of corticosteroids.

----- END OF PAPER -----

Course : B. Pharmacy
Subject Name : Pharmacology-II

Sem: V
Subject Code
:BP503T
Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Explain electro-physiology of heart.
- ii) Classify anti arrhythmic drugs.
- iii) Explain mode of action of Nitroglycerine.
- iv) Explain therapeutic uses of anti coagulants.
- v) Define and classify autacoids.
- vi) Explain role of growth hormone and enlist Growth hormone inhibitors.
- vii) Explain uses of Vitamin D.
- viii) Classify Non-Steroidal Anti-Inflammatory Drugs.
- ix) Explain action of Prolactin inhibitor-Bromocriptine
- x) Enlist fibrinolytics.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Explain in detail Etiology, drug classification, Pharmacology of antihypertensive drugs.
- ii) Discuss Synthesis, storage, role and pharmacology of thyroid hormones
- iii) Explain Principle and types of Bioassay. Discuss insulin bioassay.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

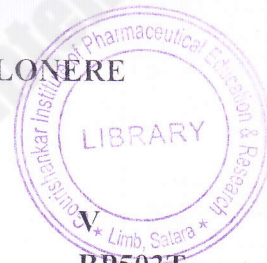
- i) Discuss mechanism of action of Digitalis
- ii) Explain pharmacology of Hematinics
- iii) Discuss Pharmacology of diuretics
- iv) Explain histamine their physiological role and classify anti-histaminics.
- v) Explain Pharmacology of Anti gout drugs
- vi) Discuss insulin preparations and oral hypoglycemic agents
- vii) Describe role of estrogen and progesterone as oral contraceptive drugs.
- viii) Discuss anti-hyperlipidemic drugs
- ix) Explain action of Prostaglandins

-----END OF THE PAPER-----

Date: 13/02/2023
25/7/2023

Course: B. Pharmacy
Subject Name: Pharmacology-II
Max Marks: 75

Sem:
Subject Code: BP503T
Duration: 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

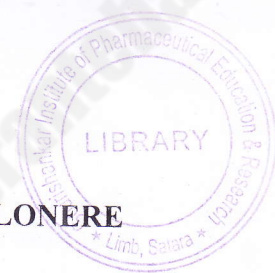
- i) Define Shock and explain how Shock can be managed.
- ii) What are Autocoids? Classify them.
- iii) Classify Anti-Arrhythmic drugs with suitable examples.
- iv) Brief note on Digoxin toxicity and its management.
- v) Give the Biosynthesis & Physiological role of Prostaglandins.
- vi) Define CHF. Write treatment measures of CHF.
- vii) What is Plasma Volume Expanders?
- viii) Classify Anti-Rheumatic drugs with suitable examples.
- ix) Enlist the hormones secreted by the Anterior and Posterior Pituitary gland.
- x) Write the MOA of Androgens used in Erectile dysfunctioning.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Enumerate the Classification of Anti-hypertensive agents with suitable examples. Write in brief the MOA of ACE inhibitors. Illustrate non-pharmacological management of Hypertension.
- ii) Write the biosynthesis and release of Thyroid hormones. Classify Anti-thyroid drugs with examples and explain their uses.
- iii) Classify oral anti-diabetic with suitable examples and explain the pharmacology of Sulfonylureas, DPP4- inhibitors.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Classify and enumerate drugs used in CCF and explain the pharmacology of Digoxin?
- ii) What are thrombolytics? Classify them with examples. Add a note on Aspirin.
- iii) Classify Anti- hyper-lipidaemics with suitable examples and explain the MOA of Statins & Bile acid sequestrants
- iv) Explain the synthesis, release and pharmacology of Histamine
- v) Classify diuretics with suitable examples. Explain the MOA of High ceiling diuretics.
- vi) Classify Anti-anginal drugs with suitable examples. Write the MOA of Organic Nitrate.
- vii) Define Bioassay. Write principle of Bioassay, application and methods.
- viii) Enlist various Histamine receptors and its antagonists. Brief the triple response of histamine.
- ix) Describe Bioassay of Insulin.



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination- Winter 2023

Date: 08/01/2024

Course: B. Pharmacy

Semester: V

Subject Name: Pharmacology- II

Subject code: BP503T

Max Marks: 75

Duration: 3 Hr.

Instructions to candidate:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Illustrate your answers with diagrams wherever necessary.

Q.1. Objective Type Questions (Answer all the questions)

(10×2)=20

- i. Define arrhythmia. Write causes of arrhythmia
- ii. Explain Biosynthesis and physiological role of 5HT
- iii. Explain MOA of DPP4-inhibitors.
- iv. Explain why potassium sparing diuretics given with loop diuretics.
- v. Explain different classes of angina.
- vi. Classify NSAID with suitable examples.
- vii. Write the principles of Bioassay.
- viii. Enlist hormone secreted by pituitary gland.
- ix. Explain MOA of Aspirin as a antiplatelet agent
- x. Classify antihypertensive agents.

Q.2. Long Answers (Answer 2 out of 3)

(2×10)= 20

- i. Explain Pathophysiology of CHF. Classify the drug's use in treatment of CHF.
Explain Pharmacology of digitalis.
- ii. What are Autocoids? Classify them. Explain synthesis, release and pharmacology of Histamine. Classify anti-histaminic drugs and write MOA, pharmacokinetics, AE of H₁ receptor antagonists.
- iii. Classify oral anti-diabetic drugs with examples and explain the MOA, Pharmacokinetics, AE of Biguanides and PPAR gamma activators.

Q.3. Short Answers (Answer 7 out of 9)

(7×5)= 35

- i. Write a note on Plasma Volume Expander.
- ii. Classify Antigout drugs. Explain MOA, Pharmacokinetics, AE of Allopurinol.
- iii. Classify fibrinolytic agents. Add and note on Streptokinase.
- iv. Write a note on SERM.
- v. Define bioassay. Explain bioassay of oxytocin.
- vi. Classify diuretics. Explain MOA and AE of loop diuretics.
- vii. Write note Thyroid hormones analogues and their inhibitors.
- viii. Classify Anti-hyperlipidemic drugs. Explain MOA of statins and Bile acid sequestrants.
- ix. Explain shock and give clinical classification of shock.

----- **End of Paper** -----

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions)

(10 X 2) = 20

- i) Describe the MOA of Cardiac glycosides used in CHF.
- ii) Write in short drugs for shock therapy.
- iii) Explain role of Dextran.
- iv) Write the applications of Anabolic steroids.
- v) Explain the biosynthesis and role of ACTH.
- vi) Write about Uterine stimulants and relaxants.
- vii) What are physiological roles of Growth Hormone?
- viii) Write the biosynthesis of Bradykinin and Kallidin.
- ix) Explain pharmacology of Vasopressin as Anti-diuretic agent.
- x) Write MOA of Vitamin K precursor drugs as coagulant.

Q.2. Long Answers (Answer 2 out of 3)

(10 X 2) = 20

- i) Classify Diuretics. Explain in detail the pharmacology of "Potassium Sparing Diuretics" class drugs.
- ii) Classify NSAID's. Explain the pharmacology of "Selective COX-2 Inhibitors" class drugs.
- iii) Classify Anti-anginal drugs. Write the pharmacology of "Nitrates" class drugs.

Q.3. Short Answers (Answer 7 out of 9)

(5 X 7) = 35

- i) Classify Anti-arrhythmic drugs. Write pharmacology of Calcium Channel Blockers.
- ii) Write the pharmacology of "Oral Anticoagulant Drugs".
- iii) Enlist Anti-Platelet drugs. Explain pharmacology of any two drugs.
- iv) Write the biosynthesis, release and physiological role of Serotonin.
- v) Write the role of Angiotensin in RAS system. Describe its physiological role and actions.
- vi) Write the biosynthesis, release and physiological role of Insulin.
- vii) What are Gonads? Explain physiological role of Progesterone and Estrogens.
- viii) Write note on Oral contraceptives.
- ix) Write the principles and applications of bioassay.

*** END OF THE PAPER ***